

**[0:00–0:45]**

Good morning. // My name is Martin Maxa, from the Department of Measurement – here at this faculty. // I worked on this together with Jakub Svatoš, from the same department. //

The subject is the acoustic detection, classification and localization of impulsive sounds, with gunshots as the target class. // The core of the contribution is a comparison of cepstral feature extractors and two classifiers for gunshot classification; localization is covered as the second function of the same sensor network. //

(// = pause. [CLICK] = advance the slide.)

# Acoustic Detection of Impulsive Sounds

## └ Motivation

## └ Motivation and research questions

### Motivation and research questions

- Acoustic detection of gunfire complements camera-based surveillance: the acoustic channel is omnidirectional and does not require line of sight.
- Required functions: detection of the impulsive event, **classification** (ideally down to calibre), and **localization** of the source – in outdoor environments that are noisy and reverberant.

### This study

1. Which cepstral representation separates gunshots from acoustically similar false alarms most reliably?
2. Neural network or support vector machine on those features?
3. How does the analysis frame length interact with environmental reflections?

[0:45–1:50]

This work is part of the development of an acoustic sensor unit for the protection of soft targets. //

Acoustic detection of gunfire complements camera-based surveillance. The acoustic channel has two properties cameras lack: it is omnidirectional, and it does not require line of sight. //

**[CUT IF LATE]** The system has to provide three functions – detection of the impulsive event, classification, ideally down to calibre, and localization of the source – in outdoor environments that are both noisy and reverberant. //

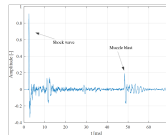
This study addresses three questions. Which cepstral representation separates gunshots from acoustically similar false alarms most reliably. // Whether a neural network or a support vector machine performs better on those features. // And how the analysis frame length interacts with environmental reflections. //

## Acoustic Detection of Impulsive Sounds

└ Signal

└ The acoustic signature of a gunshot

The acoustic signature of a gunshot



One or two components, depending on projectile speed:

- **Shock wave** ("N-wave") – supersonic only ( $v > c$ ); generated along the trajectory, as it arrives first.
- **Muzzle blast** – always present; propagates away as the bullet leaves; here  $\sim 2$  ms later.
- **Reverberations** and **reflections**.

**[1:50–3:00] (1/2)**

Both detection and localization depend on what the signal actually looks like, so let us start there. //

A gunshot recording contains one or two components, depending on the speed of the projectile. A subsonic shot produces the muzzle blast alone; a supersonic projectile adds a shock wave ahead of it. // The recording shown here is a supersonic shot, so both are present. //

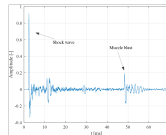
The first, at approximately two and a half milliseconds, is the shock wave. **[CUT IF LATE]** It is generated along the trajectory as a Mach cone, and the portion that reaches the microphone has the characteristic N-shaped pressure profile. **[//CUT]** //

## Acoustic Detection of Impulsive Sounds

└ Signal

└ The acoustic signature of a gunshot

The acoustic signature of a gunshot



One or two components, depending on projectile speed:

- **Shock wave** ("Nouveau") – supersonic only ( $v > c$ ); generated along the trajectory, as it arrives first.
- **Muzzle blast** – always present; propellant gases at the barrel, >2ms; here <0.5ms visible.
- **Reverberations** and after: reflections.

[1:50–3:00] (2/2)

[CLICK]

The muzzle blast – in this recording about forty-five milliseconds later – is the expansion of propellant gases at the barrel. This component is present for every shot, subsonic or supersonic. It originates at the instant of firing but propagates at the speed of sound, which is why it arrives after the shock wave. Its duration is one to three milliseconds. //

The signal between and after the two components is not part of the source signature; it consists of reflections from the surroundings. This is relevant to the choice of frame length, and I return to it in the results. //

# Acoustic Detection of Impulsive Sounds

└ Signal

└ Single-domain analysis is insufficient

Single-domain analysis is insufficient

- Spectral analysis alone often captures environmental noise rather than the gunshot itself.
- Time-domain patterns get distorted by nonlinear dispersion and obstacles.

## Consequence

Effective feature extraction must integrate both time and frequency domain processing.

**[3:00–3:30]**

Neither signal domain is sufficient on its own. //

Spectral analysis of a frame tends to characterise the acoustic environment rather than the source. // The time-domain pattern is distorted by nonlinear dispersion during propagation and by obstacles in the path. //

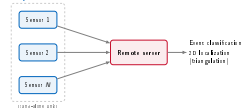
Feature extraction therefore has to combine time- and frequency-domain processing – which is what a cepstral representation does. //

## Acoustic Detection of Impulsive Sounds

└ System

└ Proposed system architecture

Proposed system architecture



- Each sensor continuously monitors its surroundings and transmits detected signals to the server for advanced processing.
- Multiple sensors provide broader coverage and adaptability to environments.
- Tuned with various firearm calibers and ammunition.

**[3:30–4:20]**

The system architecture is a distributed network of stand-alone sensor units and a remote server. //

Each unit monitors its surroundings continuously; on detecting an impulsive event it transmits the signal segment to the server. // The server performs two functions: classification of the event, and estimation of its position in two dimensions from the data of several units. //

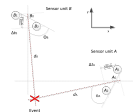
Multiple units serve two purposes. Spatial coverage – and localization, because a two-dimensional position requires at least two bearings. That is the subject of the next slide. //

The system has been tested with several firearm calibers and ammunition types. //

## Acoustic Detection of Impulsive Sounds

└ System

└ Localization – bearing from TDoA



$$\Theta = \arcsin\left(\frac{c \Delta t}{d}\right)$$

- Two MEMS microphones per unit  
[ $d = 0.38 \text{ m}$ ,  $f_s = 44.1 \text{ kHz}$ ] gives bearing  $\Theta$ .
- Two units, one bearing – they intersect at the source.
- Measured: cross-correlation with parabolic interpolation, **MAE**  $0.27^\circ$  over  $7.8 \text{ events}$  in  $[-45^\circ \text{ to } +88^\circ]$ .

**[4:20–5:20] (1/2)**

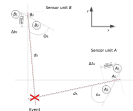
Localization. // Each sensor unit carries two MEMS microphones, almost twenty centimetres apart, sampled at 44.1 kilohertz. //

The time difference of arrival between the two channels – at most about twenty-four samples for this geometry – gives the bearing theta to the source through the relation shown: the arcsine of the speed of sound times the delay in samples, over the microphone spacing times the sampling frequency. //

## Acoustic Detection of Impulsive Sounds

└ System

└ Localization – bearing from TDoA



$$\Theta = \arcsin\left(\frac{c \Delta t}{d_B}\right)$$

- Two MEMS microphones per unit  
[ $d = 9.38 \text{ m}$ ,  $f_c = 44.1 \text{ kHz}$ ] give  
a bearing  $\Theta$ .
- Two units, one bearing – they  
intersect at the source.
- Measured: cross-correlation with  
parabolic interpolation, **MAE**  
 $0.77^\circ$  over  $7.8$  events in  $[-45^\circ$  to  
 $+88^\circ]$ .

[4:20–5:20] (2/2)

[CLICK]

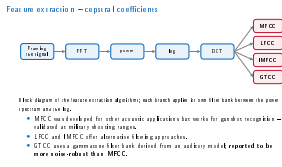
A single unit thus yields a direction. Two units yield two bearings, and their intersection is the position estimate. //

Bearing accuracy was evaluated on seventy recorded events at nine reference angles between minus forty-five and plus sixty degrees. Cross-correlation with parabolic sub-sample interpolation gives a mean absolute error of 0.77 degrees. //

# Acoustic Detection of Impulsive Sounds

## Features

### Feature extraction – cepstral coefficients



[5:20–6:25]

Classification. // All four feature extractors share the cepstral processing chain: framing, FFT, power spectrum, logarithm, discrete cosine transform. //

They differ in the filter bank applied to the power spectrum before the logarithm – which is not shown in the diagram. //

MFCC uses the mel scale. It was developed for speech, but has been validated for gunshot recognition at military shooting ranges. // LFCC uses linearly spaced filters; IMFCC an inverted mel scale. // GTCC uses a gammatone filter bank derived from a model of the auditory periphery, and is reported to be more robust to noise than MFCC. //

The experimental grid is four extractors, three frame lengths – fifteen, thirty and fifty milliseconds – and two classifiers: a neural network and a support vector machine. //

### Classification results – 9 mm class

| Frame | Extractor | ACC (%)     |             | RLC (%)     |             | MCC (%)     |            |
|-------|-----------|-------------|-------------|-------------|-------------|-------------|------------|
|       |           | SVM         | NM          | SVM         | NM          | SVM         | NM         |
| 15 ms | MFC       | 0.80        | 0.85        | 0.85        | 0.74        | 0.82        | 0.8        |
|       | LFCC      | 0.81        | 0.80        | 0.85        | 0.62        | 0.82        | 0.85       |
|       | IMFCC     | 0.58        | 0.68        | 0.87        | 0.10        | 0.68        | 0.7        |
|       | GTCC      | <b>0.82</b> | <b>0.87</b> | <b>0.78</b> | <b>0.55</b> | <b>0.82</b> | <b>0.8</b> |
| 30 ms | MFC       | 0.75        | 0.78        | 0.74        | 0.28        | 0.68        | 0.7        |
|       | LFCC      | 0.72        | 0.80        | 0.87        | 0.07        | 0.68        | 0.55       |
|       | IMFCC     | 0.58        | 0.68        | 0.87        | 0.10        | 0.68        | 0.7        |
|       | GTCC      | <b>0.74</b> | <b>0.83</b> | <b>0.72</b> | <b>0.35</b> | <b>0.68</b> | <b>0.8</b> |
| 50 ms | MFC       | 0.75        | 0.82        | 0.48        | 0.08        | 0.68        | 0.7        |
|       | LFCC      | 0.81        | 0.84        | 0.75        | 0.23        | 0.73        | 0.8        |
|       | IMFCC     | 0.82        | 0.80        | 0.78        | 0.05        | 0.73        | 0.7        |
|       | GTCC      | <b>0.74</b> | <b>0.83</b> | <b>0.82</b> | <b>0.77</b> | <b>0.83</b> | <b>0.8</b> |

Clonidine was given for 10 days. The patients: 110 women, 31 of them were

**[6:25–7:35] (1/2)**

Table one gives the results for the nine-millimetre class. //

Rows are grouped by frame length, with the four extractors within each group. // Columns are accuracy, recall, and the Matthews correlation coefficient – each for the SVM and for the neural network. //

The Matthews coefficient condenses the whole confusion matrix into one value between minus one and plus one; unlike accuracy it is not inflated by unbalanced classes, which is why it is reported here. //

## Acoustic Detection of Impulsive Sounds

## Results

## Classification results – 9 mm class

Classification results – 9 mm class

| Frame | Extractor | ACC (%) |       | REC (%) |       | MCC (+) |      |
|-------|-----------|---------|-------|---------|-------|---------|------|
|       |           | SVM     | NN    | SVM     | NN    | SVM     | NN   |
| 15 ms | MFCC      | 85.88   | 85.8  | 88.85   | 87.74 | 0.83    | 0.83 |
|       | LFCC      | 85.88   | 85.8  | 88.85   | 86.82 | 0.83    | 0.87 |
|       | IMFCC     | 85.88   | 86.8  | 88.87   | 87.18 | 0.88    | 0.87 |
|       | GTCC      | 93.32   | 97.48 | 97.18   | 93.55 | 0.83    | 0.83 |
| 30 ms | MFCC      | 84.75   | 87.8  | 87.74   | 87.18 | 0.88    | 0.78 |
|       | LFCC      | 87.18   | 85.8  | 78.87   | 78.87 | 0.88    | 0.88 |
|       | IMFCC     | 85.18   | 83.7  | 87.74   | 88.85 | 0.83    | 0.88 |
|       | GTCC      | 97.48   | 98.33 | 98.32   | 93.55 | 0.83    | 0.88 |
| 50 ms | MFCC      | 84.75   | 83.7  | 74.18   | 88.32 | 0.88    | 0.83 |
|       | LFCC      | 88.88   | 86.8  | 77.42   | 77.42 | 0.75    | 0.88 |
|       | IMFCC     | 88.88   | 88.8  | 74.18   | 88.85 | 0.75    | 0.78 |
|       | GTCC      | 97.48   | 98.33 | 98.32   | 98.77 | 0.83    | 0.88 |

(The 0.000000 results for the 9 mm class. Two partitions: 118 shots, 21 of them 9 mm.

**[6:25–7:35] (2/2)**

The GTCC rows are highlighted. For every frame length, and for both classifiers, GTCC gives the highest value of all three metrics within its group. //

The best configuration is GTCC with the neural network at fifty milliseconds: accuracy ninety-nine point one five percent, recall ninety-six point seven seven percent, and an MCC of zero point nine eight. //

For scale: the test partition contains one hundred and eighteen events, thirty-one of them nine-millimetre. Those figures correspond to one hundred and seventeen correct decisions out of one hundred and eighteen, and thirty of the thirty-one nine-millimetre shots recalled. //

## Acoustic Detection of Impulsive Sounds

## └ Results

## └ Discussion

- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- NN outperforms SVM – better at modelling complex patterns, especially with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental information, which can pull focus away from the gammatone bank.

Best configuration is the table

GTCC + NN at 50 ms = 99.35 % ACC, 99.77 % ROC, MCC 0.99.

[7:35–8:45] (1/2)

Three observations. //

First, MFCC, LFCC and IMFCC perform comparably. The choice between mel, linear and inverted-mel spacing has little effect. //

Second, GTCC performs consistently better – at every frame length and for both classifiers. **[CUT IF LATE]** The gammatone bank is derived from an auditory model and is known to be more noise-robust than MFCC; that is the probable explanation, but the mechanism was not isolated in this study. **[/CUT]** //

## Acoustic Detection of Impulsive Sounds

## └ Results

## └ Discussion

## Discussion

- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- NN outperforms SVM – better at modelling complex patterns, especially with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflection, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms = 88.35 % ACC, 88.77 % RLC, MCC 0.88.

**[7:35–8:45] (2/2)**

Third, the neural network is slightly ahead of the SVM, particularly at longer frames – a tendency across the study rather than a rule for every row of this table. //

Finally, longer frames improve accuracy, but a fifty-millisecond frame increasingly contains the impulse response of the surroundings rather than the source signature – the reflections seen on the recording earlier. The trend cannot be extrapolated further. //

## Acoustic Detection of Impulsive Sounds

└ Conclusion

└ Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2 D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibres and ammunition types.

Thank you – questions?

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**[8:45–9:30]**

To conclude. //

A distributed network of stand-alone acoustic sensors with a remote server performs detection, classification and two-dimensional localization of impulsive sounds. //

Cepstral coefficients are an effective feature representation for gunshot classification with both classifiers tested. // Gammatone-frequency coefficients outperform the mel-frequency family, and the neural network achieves slightly higher classification success than the SVM, particularly at longer frames. //

The results were obtained on recordings of real firearms of several calibres and ammunition types. //

Thank you. //